Following three problems are bonus problems. Each problem is worth 2 points.

(Bonus) An algorithm takes a positive integer N as input and repeatedly divides it by 2 until it becomes 0. Suppose n is the number of bits required to represent the integer N.

문제 2

0 / 2점

(2 points) Find the simplest function $f(N)$ such that $n = \Theta(f(N))$ i.e. find the asymptotic relationship between n and N.
(hint: use logarithms)

문제 3

0 / 2점

(2 points) What is the input size of this problem?

문제 4

0 / 2점

(2 points) Express the running time in terms of N.

문제 4

0 / 2점

(2 points) Express the running time in terms of N.

----- < 학생이 제출한 답안 > -----

Theta (n)

문제 5

2 / 2점

(2 points) Suppose that you are given a sequence of matrices $A_1, A_2, \ldots, A_n$ where matrix A_i has dimension $p_{i-1} \times p_i$. What is the input to this problem when analyzing its time complexity?

a. matrices $A_1, A_2, \ldots, A_n$
b. dimensions $p_0, p_1, \ldots, p_n$
c. n
d. $n + 1$

- a
- b
- c
- d

문제 6

1 / 1점

(1 points) Select the correct statement for the blank.
"In time complexity analysis, we are usually interested in the ( ) case because it gives an upper bound."

a. lucky
b. best
c. average
d. worst
e. wrong

- a
- b
- c
- d
- e

문제 7 0 / 1점

(1 points) $n^2 + 3n + 1 = O(n^2)$

- True
- False

문제 8 1 / 1점

(1 points) $5n \log n + n \log n^5 = \Theta(n \log n)$

- True
- False

문제 9 1 / 1점

(1 points) $\Theta\left(\frac{n^3+n^2+5n\cdot\log n}{100n^{0.01}+\log\left(\frac{n}{2}\right)}\right) = \Theta\left(n^{2.99}\right)$

- True
- False

문제 10

1 / 1점

(1 points) If $f(n) \in O\left(g(n)\right)$, $f(n)$ is always better(more efficient) then $g(n)$ for any n.

- True
- False

문제 11

0 / 1점

(1 points) If $f(n) \in o\left(g(n)\right)$ and $g(n) \in O\left(k(n)\right)$, $f(n) \in O\left(k(n)\right)$.

- True
- False

문제 12

0 / 3점

(3 points) Is the following statement true? Explain why?
"If an algorithm has the complexity $O(n^2 \lg n)$ and $\Omega(n^2)$, then the algorithm is optimal and its tight bound is $\Theta(n^2)$."

----- < 학생이 제출한 답안 > -----

$n^2 \lg n <= n^2$ => $0 <= c * f(n) <= c * g(n)$ => Satisfy both upper and lower bound. So its tight bound is n^2.

문제 13

3 / 3점

(3 points) By using the substitution method, prove that

$$T(n) = O(n^2)$$

$$\text{where } T(n) = \begin{cases} \Theta(1) & \text{if } n = 1 \\ 3T(n/4) + cn^2 & \text{if } n > 1 \end{cases}$$

Use hand-writing and upload your answer.

문제 14

3 / 3점

Using Mater theorem, what is T(n) in the previous problem? Show your solution process. Use hand-writing and upload your answer.

Use 3.png

문제 15

2 / 2점

What are the two things that must be checked before applying Dynamic Programming?

----- < 학생이 제출한 답안 > -----

overlapping subproblems and optimization

문제 16

1 / 2점

Explain why the brute-force approach for solving matrix parenthesization is inefficient.

----- < 학생이 제출한 답안 > -----

Because the time complexity is Big Omega notation(n^2).

문제 17

0 / 2점

Given matrices A, B, and C with their dimensions (5x15), (15x15), and (15x5) respectively, what is the number of scalar multiplications to compute the following parenthesization $(A \cdot (B \cdot C))$?

375

1,500 ((오차 허용): 0)

문제 18

2 / 2점

What is the best parenthesization for computing the matric chain multiplication A·B·C with the dimensions above to get the minimum number of scalar multiplications?

----- < 학생이 제출한 답안 > -----

It is same number of scalar multiplications both ((A * B) * C) and (A * (B * C)).

문제 19

1.5 / 3점

(3 points) Select all correct statements for the two pseudocode examples below.

```
LCS-Length(X,Y) {
  m ← length[X]; n ← length[Y]
  for i ← 0 to m  c[i,0] ← 0
  for j ← 0 to n  c[0,j] ← 0
  for i ← 1 to m
    for j ← 1 to n
      if X[i] = Y[j]
        then c[i,j] ← c[i-1,j-1]+1; b[i,j] ← '↖'
      else
        if c[i-1,j] ≥ c[i,j-1]
          then c[i,j] ← c[i-1,j]; b[i,j] ← '↑'
        else c[i,j] ← c[i,j-1]; b[i,j] ← '←'
  return c and b
}
```

```
LCS-Length(X,Y) {
  m ← length[X]; n ← length[Y]
  for i ← 0 to m  c[i,0] ← 0
  for j ← 0 to n  c[0,j] ← 0
  for i ← 1 to m
    for j ← 1 to n
      if X[i] = Y[j]
        then c[i,j] ← c[i-1,j-1]+1; b[i,j] ← '↖'
      else
        if c[i-1,j] > c[i,j-1]
          then c[i,j] ← c[i-1,j]; b[i,j] ← '↑'
        else c[i,j] ← c[i,j-1]; b[i,j] ← '←'
  return c and b
}
```

a. In the problem of "finding one LCS (Longest Common Subsequence)", both pseudocodes function correctly without any issues.
b. Both pseudocodes produce the same c table.
c. Although the b table in one pseudocode differs from the table 'b' in another, the LCS obtained by the PRINT-LCS() function in the lecture material remains the same.

- [x] a
- [ ] b
- [ ] c

Given sequences $X = (A, B, C, B, D)$, $Y = (B, A, C, D, B)$,

answer the following two questions.

문제 20

3 / 3점

(3 points) Draw the c table, where $c[i, j]$ stores the length of LCSs of X_i and Y_j.

Use 5.png

문제 21

3 / 3점

(3 points) Draw the b table for direction indicators.

Use the left algorithm of LCS-Length() in the previous problem.

Use 6.png

문제 22

1 / 1점

Please select the difficulty level of this quiz. (0: very easy, 5: very difficult)

4

0 ((오차 허용): 0)